

Q1 :- The standard beverage can holds 12 fl. oz & has a vol of 21.6 in^3 . What dimensions yield the min surface area? Find the min surface area.

Solution We want to minimize s , the surface area of a right circular cylinder, given by

$$s(h, r) = 2\pi rh + 2\pi r^2$$

subject to the volume constraint

$$\pi r^2 h = 21.66,$$

or
$$\pi r^2 h - 21.66 = 0.$$

Note that $s(h, r)$ does not have a minimum without the constraint.

1. We form the Lagrange function S , given by

$$S(h, r, \lambda) = 2\pi rh + 2\pi r^2 - \lambda(\pi r^2 h - 21.66).$$

2. We find the first partial derivatives:

$$\frac{\partial S}{\partial h} = 2\pi r - \lambda\pi r^2,$$

$$\frac{\partial S}{\partial r} = 2\pi h + 4\pi r - 2\lambda\pi rh,$$

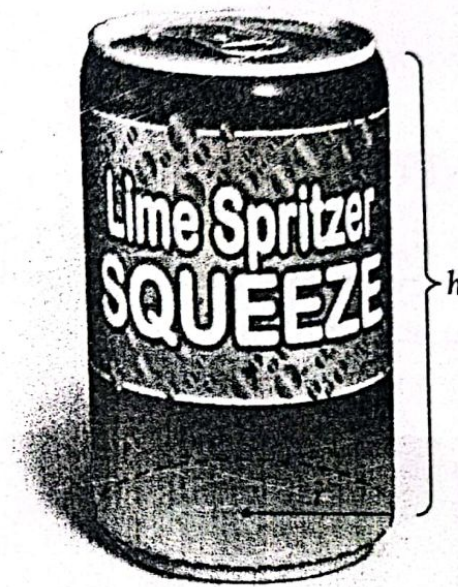
$$\frac{\partial S}{\partial \lambda} = -(\pi r^2 h - 21.66).$$

3. We set each derivative equal to 0 and solve the resulting system:

$$2\pi r - \lambda\pi r^2 = 0,$$

$$2\pi h + 4\pi r - 2\lambda\pi rh = 0,$$

$$-(\pi r^2 h - 21.66) = 0, \quad \text{or} \quad \pi r^2 h - 21.66 = 0.$$



We can solve equation (8) for r :

$$\pi r(2 - \lambda r) = 0$$

$$\pi r = 0 \quad \text{or} \quad 2 - \lambda r = 0$$

$$r = 0 \quad \text{or} \quad r = \frac{2}{\lambda}. \quad \text{We assume } \lambda \neq 0.$$

Since $r = 0$ cannot be a solution to the original problem, we solve for h by substituting $2/\lambda$ for r in equation (9):

$$2\pi h + 4\pi \cdot \frac{2}{\lambda} - 2\lambda\pi \cdot \frac{2}{\lambda} \cdot h = 0$$

$$2\pi h + \frac{8\pi}{\lambda} - 4\pi h = 0$$

$$\frac{8\pi}{\lambda} - 2\pi h = 0$$

$$\frac{8\pi}{\lambda} = 2\pi h,$$

$$\frac{4}{\lambda} = h.$$

so

Since $h = 4/\lambda$ and $r = 2/\lambda$, it follows that $h = 2r$. Substituting $2r$ for h in equation (10) yields

$$\pi r^2(2r) - 21.66 = 0$$

Using equation (1)

$$2\pi r^3 - 21.66 = 0$$

$$2\pi r^3 = 21.66$$

$$\pi r^3 = 10.83$$

$$r^3 = \frac{10.83}{\pi}$$

$$r = \sqrt[3]{\frac{10.83}{\pi}} \approx 1.51 \text{ in.}$$

Thus, when $r = 1.51$ in., we have $h = 3.02$ in. The surface area is then a minimum and is approximately

$$2\pi(1.51)(3.02) + 2\pi(1.51)^2, \quad \text{or about } 42.98 \text{ in}^2.$$

Test points can be used to support our assertion that this value is a minimum. 3 

Q2:- Each unit of a product can be made on either machine A or machine B. The nature of the machines makes their cost fns differ:

$$\text{Machine A : } C(x) = 10 + \frac{x^2}{6}$$

$$\text{Machine B : } C(y) = 200 + \frac{y^3}{9}$$

Construct the cost fn. How many units should be made on each machine in order to minimize total costs if $x+y=10,000$ units are required?

The cost production for machine A and B are $C(x) = 50 + \frac{x^2}{6}$

$$\text{and } C(y) = 260 + \frac{y^3}{9}$$

$$x + y = 12210$$

Calculation :

$$\text{Assume } C(x, y) = 50 + \frac{x^2}{6} + 260 + \frac{y^3}{9} \text{ and } f(x, y) = x + y - 12210$$

$$C(x, y) = 310 + \frac{x^2}{6} + \frac{y^3}{9} \text{ and } f(x, y) = x + y - 12210$$

Apply the method of lagranges multipliers,

$$\nabla C(x, y) = \lambda \nabla f(x, y)$$

$$\Rightarrow C_x(x, y) = \lambda f_x(x, y) \text{ and } C_y(x, y) = \lambda f_y(x, y)$$

$$\Rightarrow \frac{x}{3} = \lambda \text{ and } \frac{y^2}{3} = \lambda$$

$$\Rightarrow \frac{x}{3} = \frac{y^2}{3}$$

$$\Rightarrow x = y^2$$

$$x + y = 12210 \text{ becomes } y^2 + y = 12210$$

$$\Rightarrow y^2 + y = 12210$$

$$\Rightarrow y^2 + y - 12210 = 0$$

$$\Rightarrow y^2 + 111y - 110y - 12210 = 0$$

$$\Rightarrow y(y + 111) - 110(y + 111) = 0$$

$$\Rightarrow (y - 110)(y + 111) = 0$$

$$\Rightarrow y = -111, 110$$

Neglect $y = -111$.

$$y = 110$$

$$x = 12100$$

Machine A = 12100 units

Machine B = 110 units

Q4:- A rectangular box without a lid is to be made from 12m^2 of cardboard. Find the max vol of such a box.

Answer: Let x, y, z be the length, width, and height of such a box. The objective function that we want to maximize is the volume of this box:

$$f(x, y, z) = xyz.$$

The constraint we have is:

$$g(x, y, z) = xy + 2yz + 2zx - 12 = 0,$$

because we have to make the two pairs of side walls and one base, but we do not need the lid (top surface), which is why xy is not $2xy$ in the above equation.

Step 1: Compute all the points (x, y, z) that satisfy $\nabla f(x, y, z) = \lambda \nabla g(x, y, z)$ and the constraint $g(x, y, z) = 0$. The gradient vectors can be computed easily:

$$\nabla f(x, y, z) = \langle f_x, f_y, f_z \rangle = \langle yz, zx, xy \rangle.$$

$$\nabla g(x, y, z) = \langle g_x, g_y, g_z \rangle = \langle y + 2z, x + 2z, 2y + 2x \rangle.$$

Thus, we have the following four equations:

$$yz = \lambda(y + 2z), \quad (1)$$

$$zx = \lambda(x + 2z), \quad (2)$$

$$xy = \lambda(2y + 2x), \quad (3)$$

$$xy + 2yz + 2zx - 12 = 0. \quad (4)$$

There are four unknowns and four equations. Hence we should be able to solve these. Multiplying x to the both sides of (1) and y to those of (2) lead to

$$xyz = \lambda(xy + 2zx),$$

$$xyz = \lambda(xy + 2yz).$$

Subtracting one from the other gives us

$$0 = \lambda(2zx - 2yz) = 2\lambda z(x - y).$$

Hence, there are three possibilities, $\lambda = 0$ or $z = 0$ or $x = y$. If $\lambda = 0$, then inserting it to (1)–(3) leads to $yz = zx = xy = 0$, thus $0 + 2 \times 0 + 2 \times 0 - 12 = 0$, which leads to $-12 = 0$,

which is impossible. Thus, $\lambda \neq 0$. Also, $z = 0$ cannot happen since the height of the box must be positive. Therefore, we must have $x = y$. Inserting $x = y$ to (3) leads to

$$x^2 = 4\lambda x,$$

which leads to $x = 4\lambda$ since $x \neq 0$. Plugging $x = 4\lambda$ into (1) gives us

$$4\lambda z = \lambda(4\lambda + 2z) \implies z = 2\lambda.$$

Therefore, we have $x = y = 4\lambda$ and $z = 2\lambda$. Finally, inserting these to the constraint equation (4) leads to

$$16\lambda^2 + 4 \cdot 2\lambda \cdot 4\lambda - 12 = 0 \implies 4\lambda^2 - 1 = 0 \implies \lambda = \pm \frac{1}{2}.$$

But we must have $x > 0$, $y > 0$, $z > 0$, to be a box. Thus, we must have $\lambda = 1/2$. This leads to $x = y = 2$, $z = 1$.

Step 2: Since there is only one such point, there is no need to compare the function value at this point with that of the other point. Therefore the maximum volume of the box is

$$f(2, 2, 1) = 2 \cdot 2 \cdot 1 = 4 \text{ m}^3.$$

Note: If you want to make sure that this is the maximum volume, then you can check the volume for some other values of (x, y, z) satisfying the constraint (4). For example, $(x, y, z) = (1, 1, 11/4)$ is such a point. But it is clear that

$$f(1, 1, 11/4) = 11/4 = 2.75 < 4 = f(2, 2, 1).$$

You must be satisfied by now!

Q5:- A Computer Company has the following Cobb-Douglas production fcn for a certain product

$$P(x, y) = 800x^{3/4}y^{1/4}$$

where x is labor & y is capital, both measured in dollars.

Suppose the Company can make a total investment in labor & capital

of \$1,000,000. How should it allocate the investment b/w labor & capital in order to maximize production?

Q6:- Find the absolute max & min values of

$$f(x, y) = y^2 - x^2 - y + x$$

in the closed triangular region bounded by lines $x=0$, $y=0$ & $y = 1 - x$ ^{x -axis, y -axis}

$$P(x, y) = 800 x^{3/4} y^{1/4} \rightarrow (a)$$

$$g(x, y) = x + y - 1000000 \quad x + y - 1000000 = 0 \rightarrow (b)$$

$$F(x, y, \lambda) = 800 x^{3/4} y^{1/4} - \lambda \cdot (x + y - 1000000)$$

$$\frac{\partial F}{\partial x} = \frac{600 y^{1/4}}{x^{1/4}} - \lambda$$

$$\frac{\partial F}{\partial y} = \frac{200 x^{3/4}}{y^{3/4}} - \lambda$$

$$\frac{\partial F}{\partial \lambda} = -(x + y - 1000000)$$

$$\frac{\partial F}{\partial x} = 0$$

$$\frac{600 y^{1/4}}{x^{1/4}} - \lambda = 0$$

$$\frac{\partial F}{\partial y} = 0$$

$$\frac{200 x^{3/4}}{y^{3/4}} - \lambda = 0$$

$$\frac{600 y^{1/4}}{x^{1/4}} = \lambda \rightarrow (i)$$

$$\frac{200 x^{3/4}}{y^{3/4}} = \lambda \rightarrow (ii)$$

Equating (i) & (ii)

$$\frac{600 y^{1/4}}{x^{1/4}} = \frac{200 x^{3/4}}{y^{3/4}}$$

$$3 y^{1/4} \cdot y^{3/4} = x^{3/4} \cdot x^{1/4}$$

$$3y = x \rightarrow \text{(iii)}$$

Substituting (iii) in (b) we get

$$3y + y - 1000,000 = 0$$

$$y = \$ 250,000$$

$$x = \$ 750,000$$

So total investment for labor is \$ 750,000 & \$ 250,000 for capital to maximize production.

6. Find the absolute (global) maximum and minimum values of

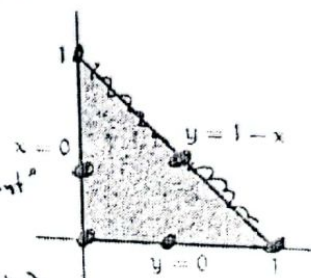
$$f(x, y) = y^2 - x^2 - y + x$$

in the closed triangular region bounded by the lines $x = 0$, $y = 0$ and $y = 1 - x$.

Interior $\nabla f = \langle 2y - 1, -2x + 1 \rangle$

$$\begin{cases} 2y - 1 = 0 \\ -2x + 1 = 0 \end{cases} \rightarrow \left(\frac{1}{2}, \frac{1}{2}\right)$$

on bdy
so "redundant"



Left ($x=0$) $f(0, y) = y^2 - y$
 $\rightarrow 2y - 1 = 0 \rightarrow y = \frac{1}{2} \quad (0, \frac{1}{2})$

Bottom ($y=0$) $f(x, 0) = -x^2 + x$
 $\rightarrow -2x + 1 = 0 \rightarrow x = \frac{1}{2} \quad (\frac{1}{2}, 0)$

Slant ($y=1-x$) $f(x, 1-x) = (1-x)^2 - x^2 - (1-x) + x$
 $= \cancel{x^2} - 2\cancel{x} + \cancel{x^2} - \cancel{x^2} + \cancel{x} + \cancel{x} + \cancel{x}$

$= 0$
 $\rightarrow$ function always 0 on $y=1-x$

Corners $(0,0), (0,1), (1,0)$

since we include corners then don't need any additional points on this side

LIST

$$f(0,0) = 0$$

$$f(0,1) = 0$$

$$f(1,0) = 0$$

$$f(0, \frac{1}{2}) = \frac{1}{4} - \frac{1}{2} = -\frac{1}{4}$$

$$f(\frac{1}{2}, 0) = -\frac{1}{4} + \frac{1}{2} = \frac{1}{4}$$

absolute max = $\frac{1}{4}$
 absolute min = $-\frac{1}{4}$